An Exploratory Analysis of the Impacts of Image Processing Techniques and Adversarial Method on Concept Bottleneck Models

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AI (Neural network) models as black box

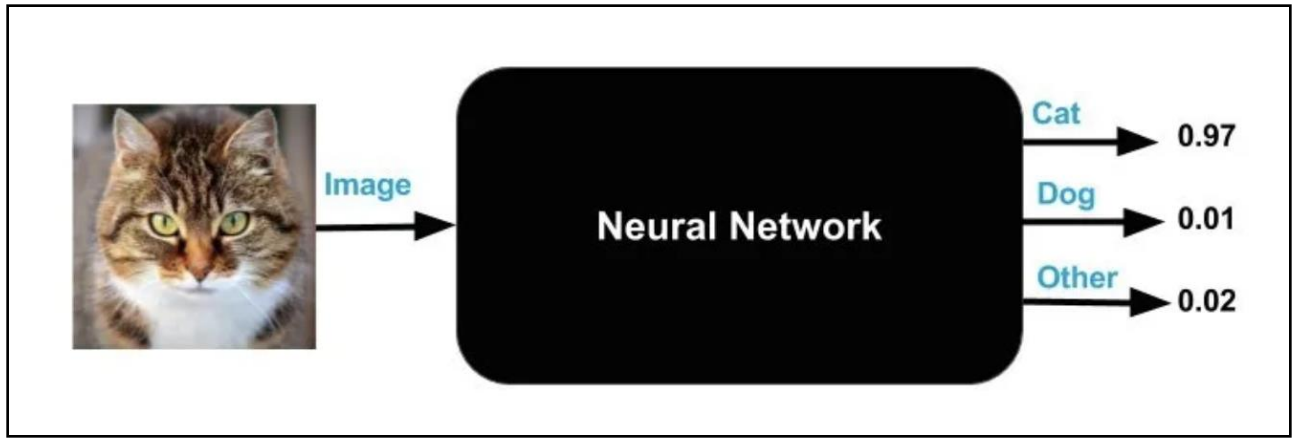
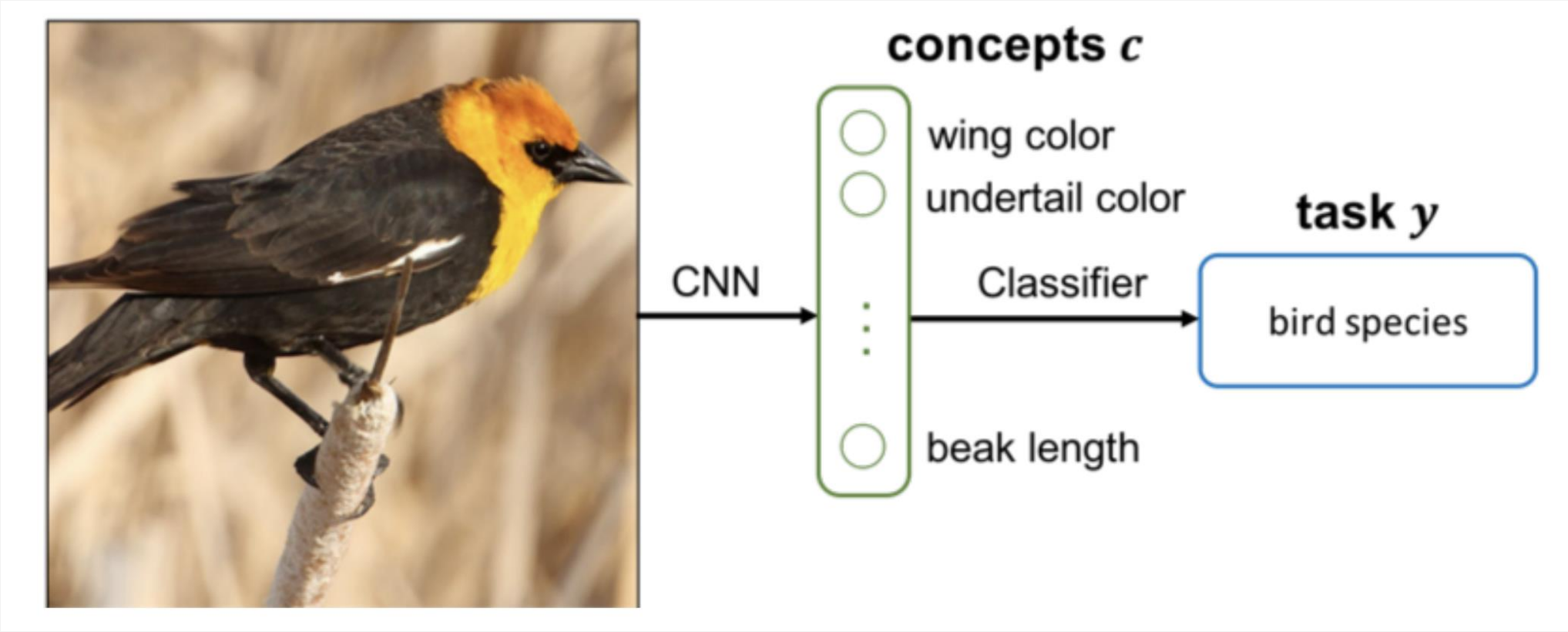


Figure: Neural networks as black box.

Image from https://medium.com/analytics-vidhya/how-to-explore-neural-networks-the-black-box-5656255d5584

Introduction to Concept Bottleneck Model

- Before predicting the actual target class, predict **concept,** hence bottleneck.
  - Concept: Human-specified attributes



Why Concept Bottleneck model ?

- **Interpretability**
  - Why certain decisions are made ?

- **Human-model interaction**
  - Allows one to intervene/change the predicted concept and propagate to the final prediction

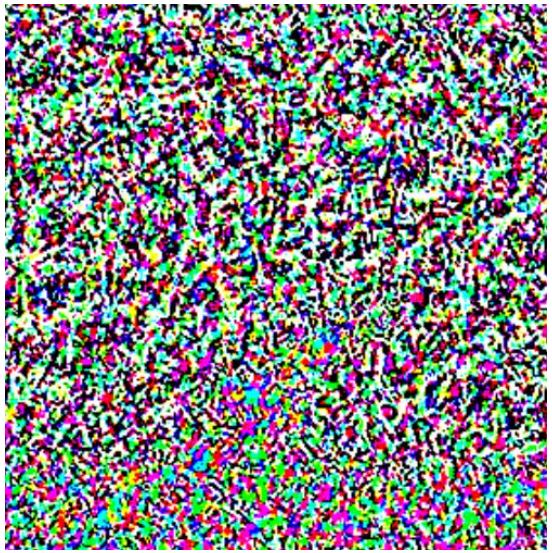
Project Motivation – Task 1: Exploring the impacts of ***image processing*** in CBM

- Intuition behind this task is to understand will the concept final prediction (class prediction/ task accuracy) gets impacted by the image processing techniques like rotation, grayscale etc.

Project Motivation – Task 2: Exploring the impacts of ***adversarial attacks*** in CBM

Black footed albatross
(Original Image)

+



$\text{sign}(\nabla_x J(\boldsymbol{\theta}, x, y))$

=

$x + \boldsymbol{\epsilon} * \text{sign}(\nabla_x J(\boldsymbol{\theta}, x, y))$
(Perturbed image with $\boldsymbol{\epsilon}$ = 0.01)

What is different from the work you referenced ?

- Exploratory analysis on CBM
- Different image processing technique impact on their performance metric i.e. Task accuracy
- Adversarial attacks impact on their performance metric.

Datasets and inputs

- CUB
  - (x,y,c) where x is of d dimension, c is of k dimension, y is real value (target, species of bird, C**lassification task**)
  - Number of target (bird species) = 200,
  - Concept= 312 before but after processing 112 binary concepts.
    - We define,
      - g(x) maps x(feature space) of d dimension into c (concept) of k dimension.
      - f(g(x)) maps concept of k dimension into final prediction.

Evaluation metrics on held-out test set

- Task Accuracy
  - How accurate f(g(x)) predicts y
- Concept Accuracy
  - How accurate g(x) predict c (average over each concept)

Different ways of learning CB

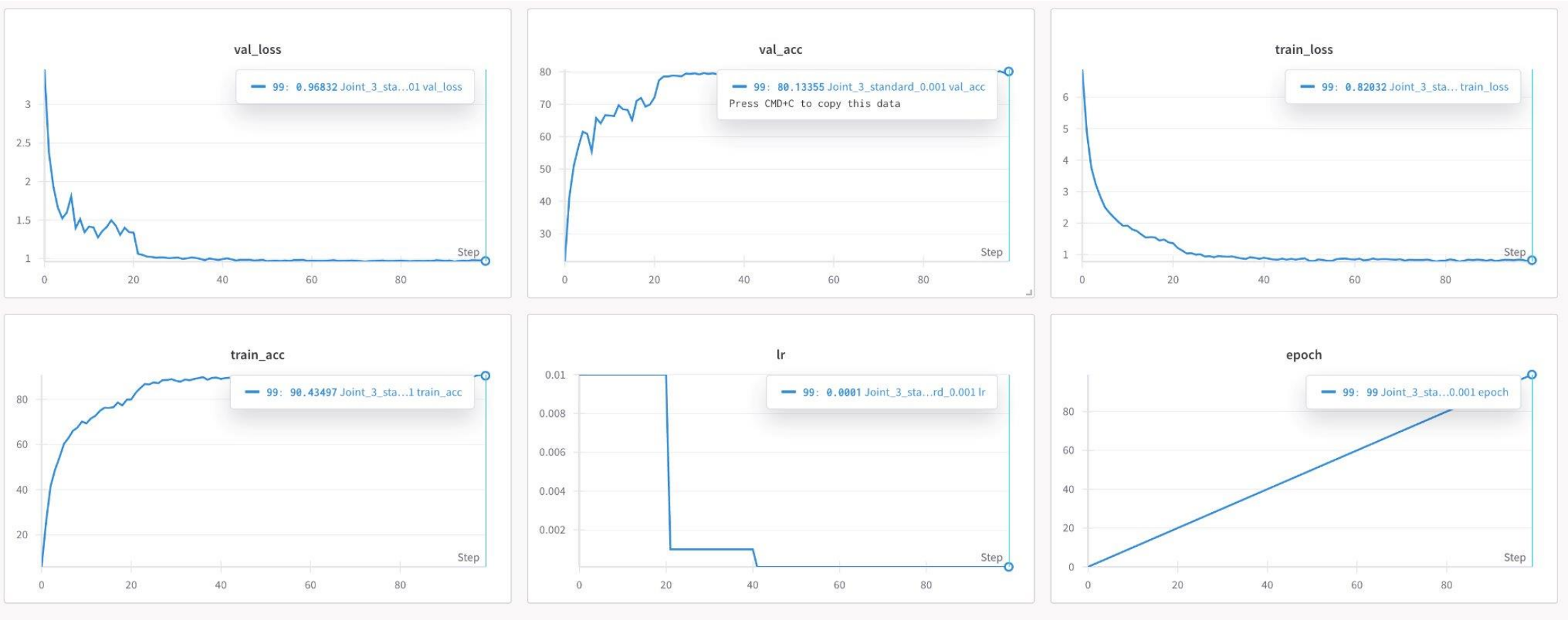
- Independent bottleneck
  - Learning f and g is independent during training we use true concept but durig testing we use predicted concept
- Sequential bottleneck
  - First learn g i.e (x -> c) and then use these predicted concept to learn f(predicted c to y)
- ***<u>Joint bottleneck</u>***
  - ***Minimizing weighted sum of both f and g***
- Standard model (like control experiment)
  - Direct x -> y

Joint model equation

$$\hat{f}, \hat{g} = \arg\min_{f,g} \sum_i \left[L_Y(f(g(x^{(i)})); y^{(i)}) + \sum_j \lambda L_{C_j}(g(x^{(i)}); c^{(i)}) \right] \text{ for some } \lambda > 0.$$

- The original paper notes joint model has ***lower task error*** compared to above models. Hence, choosing joint models here onward.
- Though paper had λ=0.01 in my case 0.001 had higher train and validation accuracy hence, using those models for creating adversarial examples and testing image processing on those models.

Joint model with λ = 0.001



Training process and models used for joint model

- Batch size = 64, Epochs = 10, 3 seeds, LR = 0.01, Optimizer = SGD
- For image to concept, InceptionV3 is used
- For concept to labels, MLP is used.
- For each seed, it took 1 hour 9 min to train in 1 NVIDIA GeForce RTX 4090 GPU.
- To create adversarial dataset, with 4 different epsilon took around 2 hours clock time.

Results with image processing techniques

Dataset	Image Processing	Task error % ↓	Concept error % ↓
	Standard	**0.1922 ± 0.0034**	**0.1669 ± 0.0053**
CUB	Grayscale	0.2929 ± 0.0026	0.1859 ± 0.0017
	Rotation	0.1843 ± 0.0045	0.1726 ± 0.0028
	Center crop	0.2774 ± 0.0023	*0.1429 ± 0.0187*
	0.5	0.1865 ± 0.0042	0.1728 ± 0.0024

Table 2: Task error and concept error where reported results are for test dataset and models are trained on provided image processing technique. Values reported are the average of 3 models on test dataset for 100 epoch with $\lambda = 0.001$ joint model. Model choosen to create the adversarial dataset is the one of 3 models. As perturbation increases, task error and concept error increases.

Results with Fast Gradient Sign Method (FGSM)

Dataset	Adversarial Attack	ϵ	Task error % $\downarrow$	Concept error % $\downarrow$
	Original (W/o attack)	-	**0.1922 ± 0.0034**	**0.1669 ± 0.0053**
CUB	FGSM	0.01	0.7357 ± 0.1619	0.1882 ± 0.0091
		0.03	0.7977 ± 0.0988	0.1942 ± 0.0059
		0.1	0.7977 ± 0.0988	0.1942 ± 0.0059
		0.5	0.9949 ± 0.0005	0.2664 ± 0.0128

Table 1: Task error and concept error with FGSM for four perturbation values. Values reported are the average of 3 models on test dataset for 100 epoch with $\lambda = 0.001$ joint model. Model choosen to create the adversarial dataset is the one of 3 models. As perturbation increases, task error and concept error increases.

Findings Summary

- Choosing different techniques during the training provides different result, in my case the transformation that was used in rotation also led to similar performance metric
- Concept bottleneck models are also vulnerable to simplest form of adversarial attacks when the model is white box.

Limitation & Future Work

- Training the model requires human-specified concepts, which might be difficult for all the data.
- Run with other adversarial attack

What lessons learned from this project ?

- Reproducing results from the proposed paper.
- Implementation and impacts of image processing techniques for CBM.
- Implementation and impact of adversarial attacks on CBM.

Thank you and QA ?

Github

Joint model with λ = 0.001

```
Average top 1 class accuracy: 80.68692
Average top 3 class accuracy: 92.07801
Average top 5 class accuracy: 95.21919
Average attribute accuracy: 82.55939
Total 1's predicted: 0.17262623896641846
Avg attribute balanced acc: 0.70316
Avg attribute F1 score: 0.53812
              precision    recall  f1-score   support

           0       0.88      0.91      0.89    515913
           1       0.59      0.50      0.54    133015

    accuracy                           0.83    648928
   macro avg       0.73      0.70      0.72    648928
weighted avg       0.82      0.83      0.82    648928
```

Seed 1

```
Average top 1 class accuracy: 80.41077
Average top 3 class accuracy: 92.71661
Average top 5 class accuracy: 95.47808
Average attribute accuracy: 83.70097
Total 1's predicted: 0.16951957690221411
Avg attribute balanced acc: 0.71786
Avg attribute F1 score: 0.56477
              precision    recall  f1-score   support

           0       0.88      0.92      0.90    515913
           1       0.62      0.52      0.56    133015

    accuracy                           0.84    648928
   macro avg       0.75      0.72      0.73    648928
weighted avg       0.83      0.84      0.83    648928
```

Seed 2

```
Average top 1 class accuracy: 81.23921
Average top 3 class accuracy: 92.25060
Average top 5 class accuracy: 95.51260
Average attribute accuracy: 83.65843
Total 1's predicted: 0.17280499531535085
Avg attribute balanced acc: 0.72018
Avg attribute F1 score: 0.56743
              precision    recall  f1-score   support

           0       0.88      0.92      0.90    515913
           1       0.62      0.52      0.57    133015

    accuracy                           0.84    648928
   macro avg       0.75      0.72      0.73    648928
weighted avg       0.83      0.84      0.83    648928
```

Seed 3

Joint model with λ = 0.001 Seed 1 Adv dataset (ε=0.01)

```
Average top 1 class accuracy: 37.86257
Average top 3 class accuracy: 64.96893
Average top 5 class accuracy: 74.39572
Average attribute accuracy: 81.23196
Total 1's predicted: 0.16555778906866614
Avg attribute balanced acc: 0.67640
Avg attribute F1 score: 0.49351
              precision    recall  f1-score   support

           0       0.86      0.91      0.88    515725
           1       0.55      0.45      0.49    132979

    accuracy                           0.81    648704
   macro avg       0.71      0.68      0.69    648704
weighted avg       0.80      0.81      0.80    648704
```

Seed 1

```
Average top 1 class accuracy: 37.89710
Average top 3 class accuracy: 63.72583
Average top 5 class accuracy: 73.67059
Average attribute accuracy: 82.26587
Total 1's predicted: 0.16210320885951066
Avg attribute balanced acc: 0.68914
Avg attribute F1 score: 0.51691
              precision    recall  f1-score   support

           0       0.87      0.92      0.89    515725
           1       0.59      0.46      0.52    132979

    accuracy                           0.82    648704
   macro avg       0.73      0.69      0.70    648704
weighted avg       0.81      0.82      0.81    648704
```

Seed 2

```
Average top 1 class accuracy: 3.53936
Average top 3 class accuracy: 18.42196
Average top 5 class accuracy: 27.69337
Average attribute accuracy: 80.03974
Total 1's predicted: 0.16568111187845305
Avg attribute balanced acc: 0.65822
Avg attribute F1 score: 0.46151
              precision    recall  f1-score   support

           0       0.86      0.90      0.88    515725
           1       0.52      0.42      0.46    132979

    accuracy                           0.80    648704
   macro avg       0.69      0.66      0.67    648704
weighted avg       0.79      0.80      0.79    648704
```

Seed 3

Joint model with λ = 0.001 Seed 1 Adv dataset (ε=0.03)

```
Average top 1 class accuracy: 27.53798
Average top 3 class accuracy: 49.03315
Average top 5 class accuracy: 58.52901
Average attribute accuracy: 80.18172
Total 1's predicted: 0.16247009421862668
Avg attribute balanced acc: 0.65750
Avg attribute F1 score: 0.46067
              precision    recall  f1-score   support

           0       0.86      0.90      0.88    515725
           1       0.52      0.41      0.46    132979

    accuracy                           0.80    648704
   macro avg       0.69      0.66      0.67    648704
weighted avg       0.79      0.80      0.79    648704
```

Seed 1

```
Average top 1 class accuracy: 26.89917
Average top 3 class accuracy: 49.77555
Average top 5 class accuracy: 59.47859
Average attribute accuracy: 81.41787
Total 1's predicted: 0.14955048835832677
Avg attribute balanced acc: 0.66477
Avg attribute F1 score: 0.47588
              precision    recall  f1-score   support

           0       0.86      0.92      0.89    515725
           1       0.56      0.41      0.48    132979

    accuracy                           0.81    648704
   macro avg       0.71      0.66      0.68    648704
weighted avg       0.80      0.81      0.80    648704
```

Seed 2

```
Average top 1 class accuracy: 6.26727
Average top 3 class accuracy: 21.21892
Average top 5 class accuracy: 30.52486
Average attribute accuracy: 80.13932
Total 1's predicted: 0.16156213003157063
Avg attribute balanced acc: 0.65602
Avg attribute F1 score: 0.45818
              precision    recall  f1-score   support

           0       0.86      0.90      0.88    515725
           1       0.52      0.41      0.46    132979

    accuracy                           0.80    648704
   macro avg       0.69      0.66      0.67    648704
weighted avg       0.79      0.80      0.79    648704
```

Seed 3

Joint model with λ = 0.001 Seed 1 Adv dataset (ε=0.1)

```
Average top 1 class accuracy: 27.53798
Average top 3 class accuracy: 49.03315
Average top 5 class accuracy: 58.54627
Average attribute accuracy: 80.18172
Total 1's predicted: 0.16246392807813734
Avg attribute balanced acc: 0.65749
Avg attribute F1 score: 0.46066
              precision    recall  f1-score   support

           0       0.86      0.90      0.88    515725
           1       0.52      0.41      0.46    132979

    accuracy                           0.80    648704
   macro avg       0.69      0.66      0.67    648704
weighted avg       0.79      0.80      0.79    648704
```

Seed 1

```
Average top 1 class accuracy: 26.89917
Average top 3 class accuracy: 49.77555
Average top 5 class accuracy: 59.47859
Average attribute accuracy: 81.41834
Total 1's predicted: 0.14953661454222572
Avg attribute balanced acc: 0.66476
Avg attribute F1 score: 0.47588
              precision    recall  f1-score   support

           0       0.86      0.92      0.89    515725
           1       0.56      0.41      0.48    132979

    accuracy                           0.81    648704
   macro avg       0.71      0.66      0.68    648704
weighted avg       0.80      0.81      0.80    648704
```

Seed 2

```
Average top 1 class accuracy: 6.26727
Average top 3 class accuracy: 21.21892
Average top 5 class accuracy: 30.54213
Average attribute accuracy: 80.13871
Total 1's predicted: 0.16156521310181532
Avg attribute balanced acc: 0.65602
Avg attribute F1 score: 0.45817
              precision    recall  f1-score   support

           0       0.86      0.90      0.88    515725
           1       0.52      0.41      0.46    132979

    accuracy                           0.80    648704
   macro avg       0.69      0.66      0.67    648704
weighted avg       0.79      0.80      0.79    648704
```

Seed 3

Joint model with λ = 0.001 Seed 1 Adv dataset (ε=0.5)

```
Average top 1 class accuracy: 0.48343
Average top 3 class accuracy: 1.43301
Average top 5 class accuracy: 2.19268
Average attribute accuracy: 71.54665
Total 1's predicted: 0.1888025971783741
Avg attribute balanced acc: 0.54887
Avg attribute F1 score: 0.27746
              precision    recall  f1-score   support

           0       0.81      0.83      0.82    515725
           1       0.29      0.27      0.28    132979

    accuracy                           0.72    648704
   macro avg       0.55      0.55      0.55    648704
weighted avg       0.71      0.72      0.71    648704
```

Seed 1

```
Average top 1 class accuracy: 0.58702
Average top 3 class accuracy: 1.15677
Average top 5 class accuracy: 2.19268
Average attribute accuracy: 74.34331
Total 1's predicted: 0.1659724620165746
Avg attribute balanced acc: 0.57110
Avg attribute F1 score: 0.30838
              precision    recall  f1-score   support

           0       0.82      0.86      0.84    515725
           1       0.34      0.28      0.31    132979

    accuracy                           0.74    648704
   macro avg       0.58      0.57      0.58    648704
weighted avg       0.72      0.74      0.73    648704
```

Seed 2

```
Average top 1 class accuracy: 0.46616
Average top 3 class accuracy: 1.39848
Average top 5 class accuracy: 2.53798
Average attribute accuracy: 74.17944
Total 1's predicted: 0.1548595353196527
Avg attribute balanced acc: 0.55853
Avg attribute F1 score: 0.28247
              precision    recall  f1-score   support

           0       0.82      0.87      0.84    515725
           1       0.33      0.25      0.28    132979

    accuracy                           0.74    648704
   macro avg       0.57      0.56      0.56    648704
weighted avg       0.72      0.74      0.73    648704
```

Seed 3

Joint model with λ = 0.001 grayscale

```
Average top 1 class accuracy: 70.34863
Average top 3 class accuracy: 86.52054
Average top 5 class accuracy: 91.18053
Average attribute accuracy: 81.18004
Total 1's predicted: 0.16362524039646925
Avg attribute balanced acc: 0.67385
Avg attribute F1 score: 0.48942
              precision    recall  f1-score   support

           0       0.86      0.91      0.88    515913
           1       0.55      0.44      0.49    133015

    accuracy                           0.81    648928
   macro avg       0.71      0.67      0.69    648928
weighted avg       0.80      0.81      0.80    648928
```

Seed 1

```
Average top 1 class accuracy: 70.91819
Average top 3 class accuracy: 86.10632
Average top 5 class accuracy: 91.24957
Average attribute accuracy: 81.60042
Total 1's predicted: 0.16989404063316732
Avg attribute balanced acc: 0.68597
Avg attribute F1 score: 0.50918
              precision    recall  f1-score   support

           0       0.87      0.91      0.89    515913
           1       0.56      0.47      0.51    133015

    accuracy                           0.82    648928
   macro avg       0.71      0.69      0.70    648928
weighted avg       0.81      0.82      0.81    648928
```

Seed 2

```
Average top 1 class accuracy: 70.86641
Average top 3 class accuracy: 86.27891
Average top 5 class accuracy: 91.11150
Average attribute accuracy: 81.45341
Total 1's predicted: 0.1709018566004241
Avg attribute balanced acc: 0.68463
Avg attribute F1 score: 0.50658
              precision    recall  f1-score   support

           0       0.87      0.90      0.89    515913
           1       0.56      0.46      0.51    133015

    accuracy                           0.81    648928
   macro avg       0.71      0.68      0.70    648928
weighted avg       0.80      0.81      0.81    648928
```

Seed 3

Joint model with λ = 0.001 rotation

```
Average top 1 class accuracy: 82.15395
Average top 3 class accuracy: 93.25164
Average top 5 class accuracy: 95.73697
Average attribute accuracy: 82.45522
Total 1's predicted: 0.16937780462547464
Avg attribute balanced acc: 0.69862
Avg attribute F1 score: 0.53133
              precision    recall  f1-score   support

           0       0.87      0.91      0.89    515913
           1       0.59      0.49      0.53    133015

    accuracy                           0.82    648928
   macro avg       0.73      0.70      0.71    648928
weighted avg       0.81      0.82      0.82    648928
```

Seed 1

```
Average top 1 class accuracy: 81.06662
Average top 3 class accuracy: 92.83742
Average top 5 class accuracy: 95.40904
Average attribute accuracy: 82.64415
Total 1's predicted: 0.17580070516297647
Avg attribute balanced acc: 0.70733
Avg attribute F1 score: 0.54420
              precision    recall  f1-score   support

           0       0.88      0.91      0.89    515913
           1       0.59      0.51      0.54    133015

    accuracy                           0.83    648928
   macro avg       0.73      0.71      0.72    648928
weighted avg       0.82      0.83      0.82    648928
```

Seed 2

```
Average top 1 class accuracy: 81.49810
Average top 3 class accuracy: 93.19986
Average top 5 class accuracy: 95.84052
Average attribute accuracy: 83.12386
Total 1's predicted: 0.17329041126288278
Avg attribute balanced acc: 0.71242
Avg attribute F1 score: 0.55386
              precision    recall  f1-score   support

           0       0.88      0.91      0.90    515913
           1       0.60      0.51      0.55    133015

    accuracy                           0.83    648928
   macro avg       0.74      0.71      0.72    648928
weighted avg       0.82      0.83      0.83    648928
```

Seed 3

Joint model with λ = 0.001 center_crop

```
Average top 1 class accuracy: 72.38522
Average top 3 class accuracy: 86.91750
Average top 5 class accuracy: 90.93890
Average attribute accuracy: 88.32875
Total 1's predicted: 0.16934852556832192
Avg attribute balanced acc: 0.78870
Avg attribute F1 score: 0.68821
              precision    recall  f1-score   support

           0       0.91      0.95      0.93    515913
           1       0.76      0.63      0.69    133015

    accuracy                           0.88    648928
   macro avg       0.83      0.79      0.81    648928
weighted avg       0.88      0.88      0.88    648928
```

Seed 1

```
Average top 1 class accuracy: 71.93649
Average top 3 class accuracy: 86.86572
Average top 5 class accuracy: 91.23231
Average attribute accuracy: 84.69876
Total 1's predicted: 0.16598913901079934
Avg attribute balanced acc: 0.72997
Avg attribute F1 score: 0.58753
              precision    recall  f1-score   support

           0       0.88      0.93      0.91    515913
           1       0.66      0.53      0.59    133015

    accuracy                           0.85    648928
   macro avg       0.77      0.73      0.75    648928
weighted avg       0.84      0.85      0.84    648928
```

Seed 2

```
Average top 1 class accuracy: 72.47152
Average top 3 class accuracy: 88.07387
Average top 5 class accuracy: 92.14705
Average attribute accuracy: 84.09038
Total 1's predicted: 0.1649135189111889
Avg attribute balanced acc: 0.71966
Avg attribute F1 score: 0.56988
              precision    recall  f1-score   support

           0       0.88      0.93      0.90    515913
           1       0.64      0.51      0.57    133015

    accuracy                           0.84    648928
   macro avg       0.76      0.72      0.74    648928
weighted avg       0.83      0.84      0.83    648928
```

Seed 3

Joint model with λ = 0.001 custom

```
Average top 1 class accuracy: 80.84225
Average top 3 class accuracy: 92.99275
Average top 5 class accuracy: 95.70245
Average attribute accuracy: 82.39358
Total 1's predicted: 0.16939937866758717
Avg attribute balanced acc: 0.69769
Avg attribute F1 score: 0.52971
              precision    recall  f1-score   support

           0       0.87      0.91      0.89    515913
           1       0.59      0.48      0.53    133015

    accuracy                           0.82    648928
   macro avg       0.73      0.70      0.71    648928
weighted avg       0.81      0.82      0.82    648928
```

Seed 1

```
Average top 1 class accuracy: 81.86054
Average top 3 class accuracy: 93.07905
Average top 5 class accuracy: 95.92682
Average attribute accuracy: 82.97007
Total 1's predicted: 0.17225485724148135
Avg attribute balanced acc: 0.70912
Avg attribute F1 score: 0.54855
              precision    recall  f1-score   support

           0       0.88      0.91      0.90    515913
           1       0.60      0.50      0.55    133015

    accuracy                           0.83    648928
   macro avg       0.74      0.71      0.72    648928
weighted avg       0.82      0.83      0.82    648928
```

Seed 2

```
Average top 1 class accuracy: 81.36002
Average top 3 class accuracy: 92.90646
Average top 5 class accuracy: 95.85778
Average attribute accuracy: 82.81119
Total 1's predicted: 0.1728758814537206
Avg attribute balanced acc: 0.70725
Avg attribute F1 score: 0.54509
              precision    recall  f1-score   support

           0       0.88      0.91      0.89    515913
           1       0.60      0.50      0.55    133015

    accuracy                           0.83    648928
   macro avg       0.74      0.71      0.72    648928
weighted avg       0.82      0.83      0.82    648928
```

Seed 3

Joint model with $\lambda = 0.001$ for rotation seed 3

Joint model with λ = 0.01

```
Average top 1 class accuracy: 72.33345
Average top 3 class accuracy: 88.16016
Average top 5 class accuracy: 92.66483
Average attribute accuracy: 87.00226
Total 1's predicted: 0.16752397800680507
Avg attribute balanced acc: 0.76670
Avg attribute F1 score: 0.65107
              precision    recall  f1-score   support

           0       0.90      0.94      0.92    515913
           1       0.72      0.59      0.65    133015

    accuracy                           0.87    648928
   macro avg       0.81      0.77      0.79    648928
weighted avg       0.86      0.87      0.86    648928
```

Seed 1

```
Average top 1 class accuracy: 72.29893
Average top 3 class accuracy: 88.36728
Average top 5 class accuracy: 92.66483
Average attribute accuracy: 87.37518
Total 1's predicted: 0.16868281226884954
Avg attribute balanced acc: 0.77347
Avg attribute F1 score: 0.66213
              precision    recall  f1-score   support

           0       0.90      0.94      0.92    515913
           1       0.73      0.60      0.66    133015

    accuracy                           0.87    648928
   macro avg       0.82      0.77      0.79    648928
weighted avg       0.87      0.87      0.87    648928
```

Seed 2

Joint model with $\lambda = 0.01$ Seed 3

Original Paper output

MODEL	y RMSE (OAI)	y ERROR (CUB)
INDEPENDENT	0.435± 0.024	0.240±0.012
SEQUENTIAL	0.418± 0.004	0.243±0.006
JOINT	0.418± 0.004	0.199±0.006
STANDARD	0.441± 0.006	0.175±0.008
NO BOTTLENECK	0.443 ± 0.008	0.173±0.003
MULTITASK	0.425± 0.010	0.162±0.002

Joint model with λ = 0.001 Seed 2

```
Average top 1 class accuracy: 80.41077
Average top 3 class accuracy: 92.71661
Average top 5 class accuracy: 95.47808
Average attribute accuracy: 83.70097
Total 1's predicted: 0.16951957690221411
Avg attribute balanced acc: 0.71786
Avg attribute F1 score: 0.56477
              precision    recall  f1-score   support

           0       0.88      0.92      0.90    515913
           1       0.62      0.52      0.56    133015

    accuracy                           0.84    648928
   macro avg       0.75      0.72      0.73    648928
weighted avg       0.83      0.84      0.83    648928
```

Joint model with λ = 0.001 Seed 3

```
Average top 1 class accuracy: 81.23921
Average top 3 class accuracy: 92.25060
Average top 5 class accuracy: 95.51260
Average attribute accuracy: 83.65843
Total 1's predicted: 0.17280499531535085
Avg attribute balanced acc: 0.72018
Avg attribute F1 score: 0.56743
              precision    recall  f1-score   support

           0       0.88      0.92      0.90    515913
           1       0.62      0.52      0.57    133015

    accuracy                           0.84    648928
   macro avg       0.75      0.72      0.73    648928
weighted avg       0.83      0.84      0.83    648928
```

Joint model with λ = 0.001 Seed 1 Adv dataset (ε=0.01)

```
Average top 1 class accuracy: 37.86257
Average top 3 class accuracy: 64.96893
Average top 5 class accuracy: 74.39572
Average attribute accuracy: 81.23196
Total 1's predicted: 0.16555778906866614
Avg attribute balanced acc: 0.67640
Avg attribute F1 score: 0.49351
              precision    recall  f1-score   support

           0       0.86      0.91      0.88    515725
           1       0.55      0.45      0.49    132979

    accuracy                           0.81    648704
   macro avg       0.71      0.68      0.69    648704
weighted avg       0.80      0.81      0.80    648704
```

Joint model with λ = 0.001 Seed 2 Adv dataset (ε=0.01)

```
Average top 1 class accuracy: 37.89710
Average top 3 class accuracy: 63.72583
Average top 5 class accuracy: 73.67059
Average attribute accuracy: 82.26587
Total 1's predicted: 0.16210320885951066
Avg attribute balanced acc: 0.68914
Avg attribute F1 score: 0.51691
              precision    recall  f1-score   support

           0       0.87      0.92      0.89    515725
           1       0.59      0.46      0.52    132979

    accuracy                           0.82    648704
   macro avg       0.73      0.69      0.70    648704
weighted avg       0.81      0.82      0.81    648704
```

Joint model with λ = 0.001 Seed 3 Adv dataset (ε=0.01)

```
Average top 1 class accuracy: 3.53936
Average top 3 class accuracy: 18.42196
Average top 5 class accuracy: 27.69337
Average attribute accuracy: 80.03974
Total 1's predicted: 0.16568111187845305
Avg attribute balanced acc: 0.65822
Avg attribute F1 score: 0.46151
              precision    recall  f1-score   support

           0       0.86      0.90      0.88    515725
           1       0.52      0.42      0.46    132979

    accuracy                           0.80    648704
   macro avg       0.69      0.66      0.67    648704
weighted avg       0.79      0.80      0.79    648704
```

Joint model with λ = 0.01 Seed 1

```
Average top 1 class accuracy: 72.33345
Average top 3 class accuracy: 88.16016
Average top 5 class accuracy: 92.66483
Average attribute accuracy: 87.00226
Total 1's predicted: 0.16752397800680507
Avg attribute balanced acc: 0.76670
Avg attribute F1 score: 0.65107
              precision    recall  f1-score   support

           0       0.90      0.94      0.92    515913
           1       0.72      0.59      0.65    133015

    accuracy                           0.87    648928
   macro avg       0.81      0.77      0.79    648928
weighted avg       0.86      0.87      0.86    648928
```

Joint model with λ = 0.01 Seed 2

```
Average top 1 class accuracy: 72.29893
Average top 3 class accuracy: 88.36728
Average top 5 class accuracy: 92.66483
Average attribute accuracy: 87.37518
Total 1's predicted: 0.16868281226884954
Avg attribute balanced acc: 0.77347
Avg attribute F1 score: 0.66213
              precision    recall  f1-score   support

           0       0.90      0.94      0.92    515913
           1       0.73      0.60      0.66    133015

    accuracy                           0.87    648928
   macro avg       0.82      0.77      0.79    648928
weighted avg       0.87      0.87      0.87    648928
```